Indian Institute of Technology Bhubaneswar

Mid Semester (Spring) Examination-2020

Subject Name: Environmental Science Technology and Management **Subject Code: ID2L003**

Date: 28-02-2023

Duration: 2 Hours

Full Marks: 40

Answer a new question on a fresh page (strictly) and all parts of a question together.
ANSWER BRIEFLY. USE BULLET POINTS. USE ILLUSTRATION WHERE NECESSARY

1. In air pollution related discussions, India though being a lower emitter is prominently shown as an important actor who needs to take urgent actions to address these global problems.
   a. Why is it that India is much more polluted than other highly emitting countries or regions such as USA or China?
   b. What are the important environmental factors that govern concentration of pollutants relevant to this discussion?
   c. Is there any other factor that could easily be controlled to reduce air pollution by government action according to you?
   d. Can individually, cities or states improve their air quality over the Indian region? Provide illustration to answer the above questions where necessary. [10]

2. If there is a looming environmental disaster due to climate change as a consequence of warming,
   a. Are there any technological fixes that could slow down these changes?
   b. How can they be generally classified based on their forcing factor? Solar Radiation CO_2 capture
   c. List their advantages and disadvantages.
   d. Which one of these fixes would you suggest as a policy maker? Provide your reasoning in bullet points. [10]

3. Answer the following,
   Check
   a. Use the relative abundance of three most important components of the environment/Earth-Climate system (e.g. air, water and land) and explain their rarity and importance to life. Use any examples or relationships that you are aware of in bullet points. [5]

4. Quantify global warming due to climate change in the past 100 years. How much is it? Are there any environmental factors that are masking this change? By how much? Why? [5]

5. What are the different global risks highlighted by World Economic Forum that is expected to slow down global economic growth? [5]

6. The inverse square law states that energy received per unit area is inversely proportional to square of the distance from the source. If we apply this concept and decide to shift earth orbit, away from sun, how much should we shift it to reduce radiative forcing equivalent to that causing global anthropogenic climate change? Use any concept on energy balance that you are aware of. Make appropriate assumptions and state it clearly. [5]